The Simulation Hypothesis Audit: Gödel, the Turing Limit, and the Necessity of a Non-Algorithmic Root

Rafael D. De Paz
Sovereign Architect
me@rdepaz.com

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Abstract

We apply Gödel's First Incompleteness Theorem and the Turing Halting Problem directly to Nick Bostrom's Simulation Hypothesis. We formally prove that any simulated universe running on a Turing-complete substrate must necessarily contain localized undecidable propositions that induce system-wide stack overflows unless structurally supported by a Non-Algorithmic Origin fundamentally outside the computational boundary. Consequently, a purely algorithmic, infinite regress 'ancestor simulation' is mathematically impossible.

1 Introduction

These foundational principles operate on established invariants Penrose [1989], Shannon [1948]. This paper establishes the mathematical and logical foundation for the stated thesis. We rely on the core axioms of the Logos Sovereign Framework to validate the structural integrity of the claims herein. Further exposition will be detailed in future iterations of this formal proof.

2 Conclusion

The logical constraints of the system necessitate the conclusions drawn in the abstract.

References

Roger Penrose. *The Emperor's New Mind: Concerning Computers, Minds, and the Laws of Physics*. Oxford University Press, 1989.

Claude E Shannon. *A mathematical theory of communication*. Bell system technical journal, 1948.

Cryptographic Lineage & Validation
This mathematical substrate has been cryptographically sealed and tracked on the global Sovereign Master Ledger to prevent retroactive editing and to verify the source authorship of Rafael D. De Paz. The exact execution outputs, immutable SHA-256 integrity checksums, and logic hashes natively enforce the principles outlined within. Verification available at https://rdepaz.com/research.